المؤسسة العامة لجسر الملك فهد
King Fahd Causeway Authority

AI-Powered Causeway Operations Copilot

Problem Framing Response

Technical Assessment Submission

Executive Summary
This proposal presents a practical AI solution that uses existing CCTV and operational data to detect incidents and congestion early, support faster operational decisions and improve communication with travellers across the King Fahd Causeway.

Prepared for	King Fahd Causeway Authority
Document type	Problem Framing Proposal
Focus	Operations, safety, and traveller experience

Proposed Solution

I propose an AI-powered Operations Copilot that uses computer vision and generative AI to support daily operations on the causeway. The system uses existing CCTV and operational data to detect incidents and congestion early, support faster decisions and provide clear updates to travellers. The goal is to move from reactive operations to a more proactive and data-driven approach.

Why this solution fits KFCA
It builds on systems that already exist, does not require additional sensors or devices and supports KFCA priorities around efficiency, safety, customer experience and sustainability.

1. Problem Definition

KFCA operates a high-volume cross-border corridor where even small issues can quickly lead to major delays. For example, a stopped vehicle or a sudden increase in traffic can cause queues to grow quickly. This leads to longer waiting times, lower throughput, higher safety risks and more pressure on the operations team. It also affects the traveller experience and increases vehicle idling.

Today, much of the monitoring depends on manual observation of CCTV and coordination between teams. This makes the response slower, especially during busy periods. In many cases, the issue is noticed after congestion has already started to build. The problem is how to detect these situations earlier, respond faster and communicate clearly with travellers and partner agencies. Solving this will improve efficiency, safety and customer experience. It also supports sustainability by reducing unnecessary delays and congestion.

2. Data Requirements

The solution depends on combining video data with operational data and internal knowledge. It is designed to use existing systems, so no additional sensors or devices are required.

First, video data comes from existing CCTV cameras across the causeway. This includes entry and exit points, inspection lanes, truck areas and key bottlenecks. This data is used to detect incidents, stopped vehicles, lane blockages, unsafe movement and queue build-up.

Second, historical data is needed to train and validate the models. This includes past footage or labelled images showing accidents, normal traffic, congestion levels and different vehicle types. This data can come from CCTV archives, incident reports and operator logs.

Third, operational data is important to make the system useful in real conditions. This includes lane status, vehicle counts, processing times and traffic patterns during different times and seasons. It helps connect what is seen on cameras with what is happening in operations.

In addition, the generative AI part requires internal documents such as SOPs, escalation procedures and approved communication templates. This ensures that recommendations and messages are accurate and aligned with KFCA policies. Finally, data should be handled securely. Access should be controlled, actions should be logged and sensitive information in video should be protected where needed.

3. Proposed Approach

The solution is built in three main parts. Each part has a clear role and supports daily operations.

1. Computer Vision Layer

This part acts as the system's eyes. It analyses live CCTV footage to understand what is happening on the ground. It can detect vehicles, trucks, pedestrians, and unusual situations. For example, it can identify a stopped vehicle, a blocked lane, or a queue that is starting to grow. It can also track movement over time to understand how traffic is flowing. Models like YOLOv8 or RT-DETR can be used for detection and tools like ByteTrack or DeepSORT help track movement. These are widely used and can work in near real time.

2. Operational Intelligence Layer

This part turns what the system sees into useful insights.

It combines the camera data with operational data such as lane status, vehicle counts and processing times. This helps estimate queue length, detect areas under pressure and predict short-term delays. For example, it can show that a specific lane will become congested in the next 20 minutes if no action is taken.

Models like XGBoost or LightGBM can be used here and time-series models can be added if enough historical data is available. The main value is helping the team act earlier instead of reacting late.

3. Generative AI Copilot

This part supports the operators directly.

Instead of just showing alerts, the system explains what is happening in simple terms and suggests what to do next. It uses internal procedures to make sure the recommendations are aligned with how KFCA operates. It can also generate short messages for travellers in Arabic and English based on the situation. This helps reduce workload and makes communication faster and more consistent.

Human in the Loop

The system supports the team but does not replace them. It detects and suggests but the operator makes the final decision. They can review, approve, or ignore the recommendations. This keeps the system practical and safe to use.

This approach helps KFCA detect issues earlier, respond faster, and manage operations more smoothly using the systems already in place.

4. Success Metrics

To measure the impact of the solution, I would focus on four main areas.

1. Safety

The first priority is improving safety and response time. We can measure how quickly incidents are detected and how accurate the system is. It is also important to track how much faster the team responds compared to today. For example, the goal would be to detect most incidents within one minute and reduce delays in response. Over time, we should also see fewer secondary issues caused by late detection.

2. Operations

The second area is operational performance. This includes tracking queue length, waiting time and how many vehicles pass through each hour. It is also important to measure how accurate the delay predictions are. A key goal would be to reduce waiting time during peak periods and improve overall traffic flow. Even a small improvement here can have a large impact.

3. Customer Experience

The third area is the traveller experience. We can measure how quickly updates are shared with travellers and how accurate those updates are. Feedback and complaints can also give a clear signal of improvement. Even when delays cannot be avoided, clear and timely communication can improve how travellers experience the journey.

4. System Performance and Adoption

Finally, we need to make sure the system is reliable and actually used. This includes system uptime, speed and how often operators rely on it. We can also track how many recommendations are accepted or used by the team. A successful system is not just accurate but also trusted and used in daily operations.

Pilot Approach

The best way to validate the solution is to start with a pilot in a few key locations.

We can compare performance before and after deployment and also compare AI-supported monitoring with manual monitoring. This helps show both operational improvement and real business impact. This way, success is measured not only by technical accuracy but by real improvements in safety, efficiency and customer experience.

5. Conclusion

The proposed AI-powered Operations Copilot provides a practical way for KFCA to improve how the causeway is managed on a daily basis. By using existing CCTV and operational data, the solution enables earlier detection of issues, faster response and clearer communication with travellers. It supports a shift from reactive monitoring to a more proactive and data-driven approach. This leads to better traffic flow, improved safety and a stronger customer experience while also supporting sustainability by reducing unnecessary delays. Overall, the solution aligns well with KFCA's priorities and can be implemented in a scalable and cost-effective way.